# Why pipeline

Production SPICE runs OP first, initializes storage elements, then steps TRAN

# The idea

- GOLDEN: OP V2=3.0; TRAN from v2_init=0 → final_v2≈2.255181

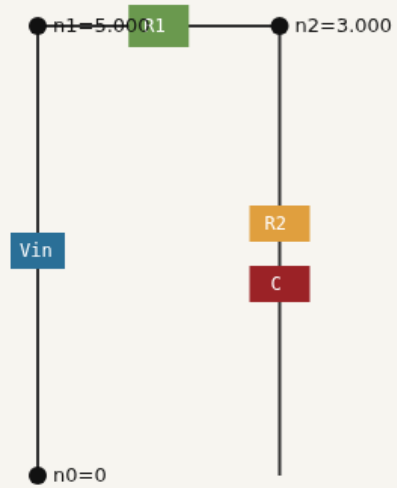
# OP

OP → TRAN PIPELINE

## OP

Step 1 / 5 · op · module04-03-op-tran-pipeline

OP




**Explanation**

Compute DC operating point.

- OP

```
V2=3
```

OP

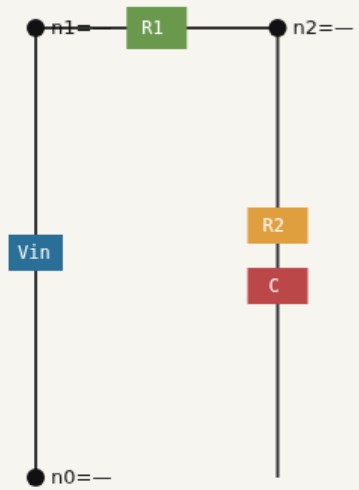
# Init C

OP→TRAN PIPELINE

## Init C

Step 2 / 5 · init · module04-03-op-tran-pipeline

elements=5




**Explanation**

Start capacitor from 0 for TRAN demo.

- init

```
0
```

Init C

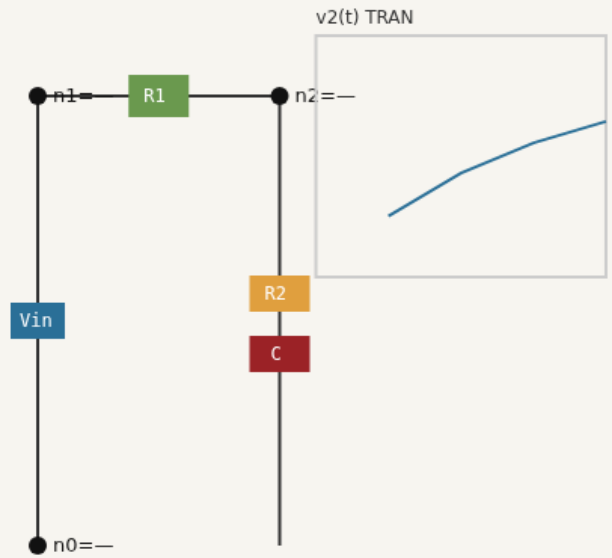
# Step loop

OP→TRAN PIPELINE

## Step loop

Step 3 / 5 · loop · module04-03-op-tran-pipeline

steps




**Explanation**

Repeat BE steps.

- loop

4

Step loop

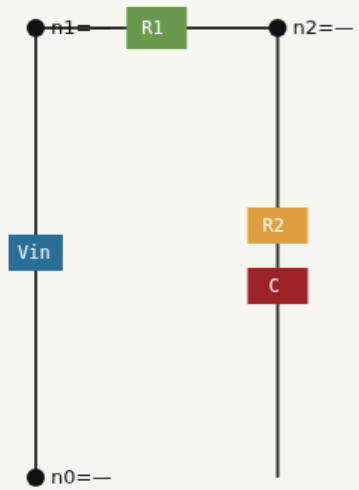
# Pipeline

OP→TRAN PIPELINE

## Pipeline

Step 4 / 5 · pipe · module04-03-op-tran-pipeline

elements=5




**Explanation**

OP then TRAN is the classic flow.

- flow

```
OP→TRAN
```

Pipeline

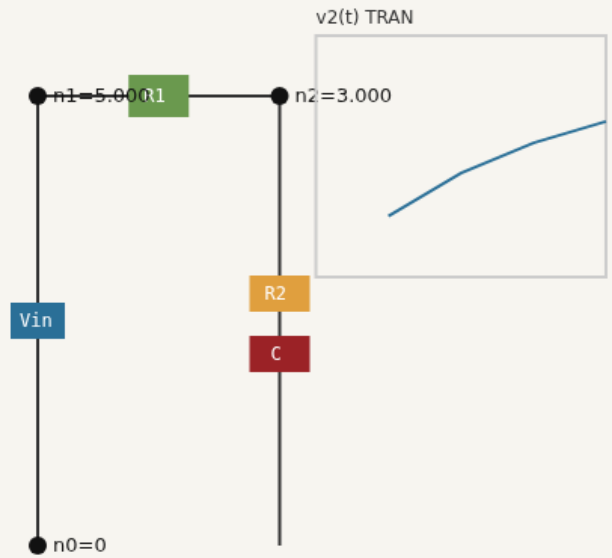
# Report

OP→TRAN PIPELINE

## Report

Step 5 / 5 · rep · module04-03-op-tran-pipeline

dc=3.0000000000000004 tran=2.255




**Explanation**

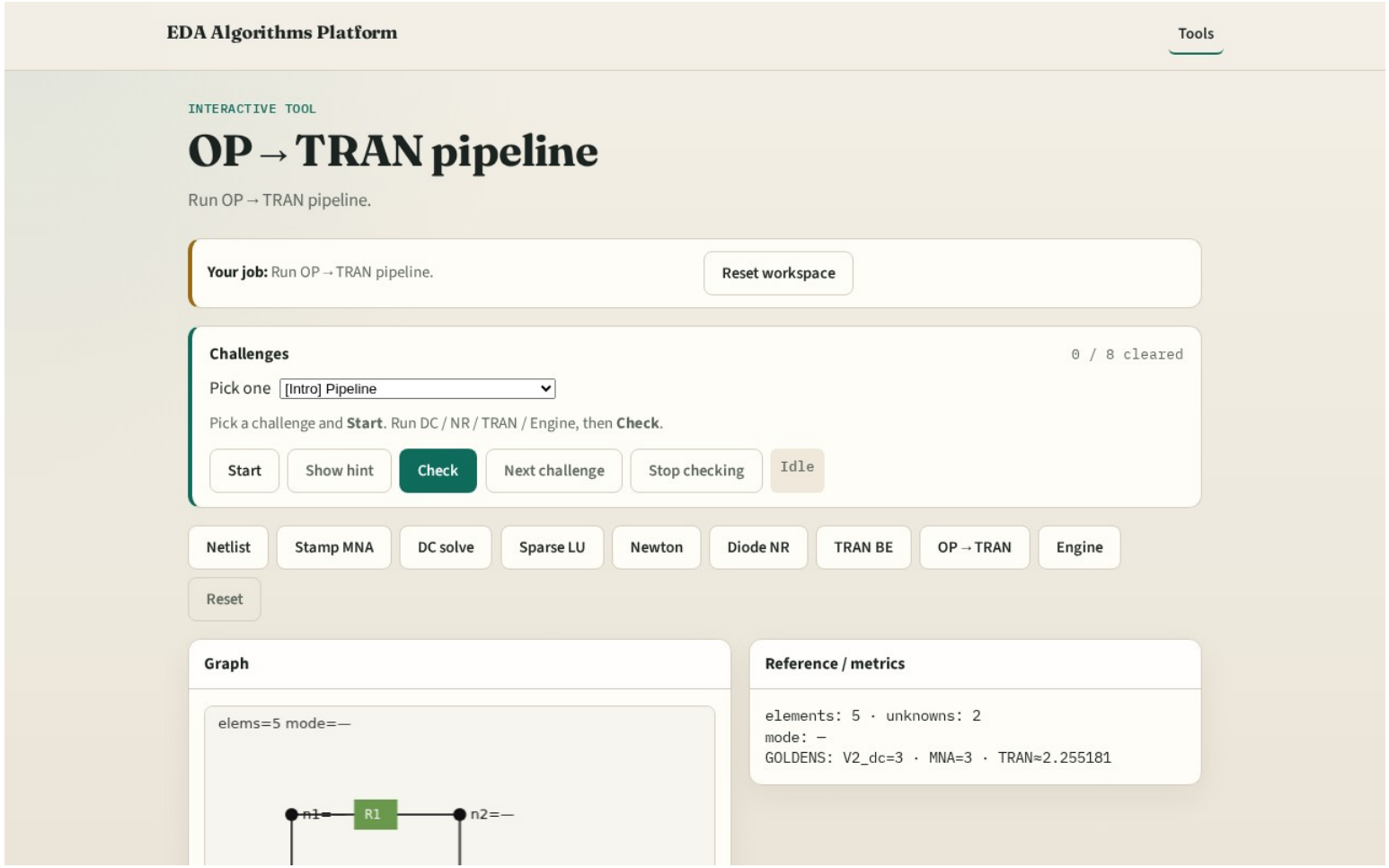
Report DC and TRAN endpoints.

- report

```
metrics
```

Report

# Browser lab track




Browser lab starter

# Implement track

- Call dc_linear_solve() then transient_be(v2_init=0.0); report both v2 values

# Pitfalls

- Starting TRAN from OP cap voltage when lab sets v2_init=0
- Skipping OP check

# Your turn

- Next: full spice_engine integration